

# CLUTRR: A Diagnostic Benchmark for Inductive Reasoning from Text

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## Abstract

The recent success of natural language understanding (NLU) systems has been troubled by results highlighting the failure of these models to generalize in a systematic and robust way. In this work, we introduce a diagnostic benchmark suite, named CLUTRR, to clarify some key issues related to the robustness and systematicity of NLU systems. Motivated by classic work on inductive logic programming, CLUTRR requires that an NLU system infer kinship relations between characters in short stories. Successful performance on this task requires both extracting relationships between entities, as well as inferring the logical rules governing these relationships. CLUTRR allows us to precisely measure a model’s ability for systematic generalization by evaluating on held-out combinations of logical rules, and it allows us to evaluate a model’s robustness by adding curated noise facts. Our empirical results highlight a substantial performance gap between state-of-the-art NLU models (e.g., BERT and MAC) and a graph neural network model that works directly with symbolic inputs—with the graph-based model exhibiting both stronger generalization and greater robustness.

## 1 Introduction

Natural language understanding (NLU) systems have been extremely successful at reading comprehension tasks, such as question answering (QA) and natural language inference (NLI). An array of existing datasets are available for these tasks. This includes datasets that test a system’s ability to extract factual answers from text (Rajpurkar et al., 2016; Nguyen et al., 2016; Trischler et al., 2016; Mostafazadeh et al., 2016; Su et al., 2016), as well as datasets that emphasize commonsense inference, such as entailment between sentences (Bowman et al., 2015; Williams et al., 2018).

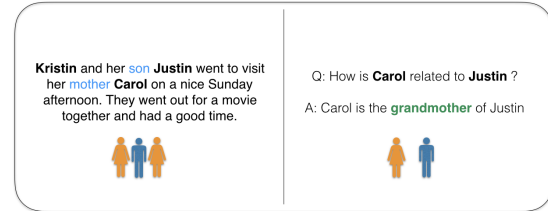


Figure 1: CLUTRR inductive reasoning task.

However, there are growing concerns regarding the ability of NLU systems—and neural networks more generally—to generalize in a systematic and robust way (Bahdanau et al., 2019; Lake and Baroni, 2018; Johnson et al., 2017). For instance, recent work has highlighted the brittleness of NLU systems to adversarial examples (Jia and Liang, 2017), as well as the fact that NLU models tend to exploit statistical artifacts in datasets, rather than exhibiting true reasoning and generalization capabilities (Gururangan et al., 2018; Kaushik and Lip-ton, 2018). These findings have also dovetailed with the recent dominance of large pre-trained language models, such as BERT, on NLU benchmarks (Devlin et al., 2018; Peters et al., 2018), which suggest that the primary difficulty in these datasets is incorporating the statistics of the natural language, rather than reasoning.

An important challenge is thus to develop NLU benchmarks that can precisely test a model’s capability for robust and systematic generalization. Ideally, we want language understanding systems that can not only answer questions and draw inferences from text, but that can also do so in a systematic, logical, and robust way. While such reasoning capabilities are certainly required for many existing NLU tasks, most datasets combine several challenges of language understanding into one, such as co-reference/entity resolution, incorporating world knowledge, and semantic parsing—making it difficult to isolate and diagnose a model’s capabilities for systematic generalization and robustness.

**Our work.** Inspired by the classic AI challenge of inductive logic programming (Quinlan, 1990)—as well as the recently developed CLEVR dataset for visual reasoning (Johnson et al., 2017)—we propose a semi-synthetic benchmark designed to explicitly test an NLU model’s ability for systematic and robust logical generalization.

Our benchmark suite—termed CLUTRR (Compositional Language Understanding and Text-based Relational Reasoning)—contains a large set of semi-synthetic stories involving hypothetical families. Given a story, the goal is to infer the relationship between two family members, whose relationship is not explicitly mentioned (Figure 1). To solve this task, a learning agent must extract the relationships mentioned in the text, induce the logical rules governing the kinship relationships (e.g., the transitivity of the sibling relation), and use these rules to infer the relationship between a given pair of entities. Crucially, the CLUTRR benchmark allows us to test a learning agent’s ability for *systematic generalization* by testing on stories that contain unseen combinations of logical rules. CLUTRR also allows us to precisely test for the various forms of *model robustness* by adding different kinds of superfluous *noise facts* to the stories.

We compare the performance of several state-of-the-art NLU systems on this task—including Relation Networks (Santoro et al., 2017), Compositional Attention Networks (Hudson and Manning, 2018) and BERT (Devlin et al., 2018). We find that the generalization ability of these NLU systems is substantially below that of a Graph Attention Network (Veličković et al., 2018), which is given direct access to symbolic representations of the stories. Moreover, we find that the robustness of the NLU systems generally does not improve by training on noisy data, whereas the GAT model is able to effectively learn robust reasoning strategies by training on noisy examples. Both of these results highlight important open challenges for closing the gap between machine reasoning models that work with unstructured text and models that are given access to more structured input.

## 2 Related Work

We draw inspiration from the classic work on inductive logic programming (ILP), a long line of reading comprehension benchmarks in NLP, as well as work combining language and knowledge graphs.

**Reading comprehension benchmarks.** Many datasets have been proposed to test the reading comprehension ability of NLP systems. This includes the SQuAD (Rajpurkar et al., 2016), NewsQA (Trischler et al., 2016), and MCTest (Richardson et al., 2013) benchmarks that focus on factual questions; the SNLI (Bowman et al., 2015) and MultiNLI (Williams et al., 2018) benchmarks for sentence understanding; and the bABI tasks (Weston et al., 2015), to name a few. Our primary contribution to this line of work is the development of a carefully designed *diagnostic* benchmark to evaluate model robustness and systematic generalization in the context of NLU.

### Question-answering with knowledge graphs.

Our work is also related to the domain of question answering and reasoning in knowledge graphs (Das et al., 2018; Xiong et al., 2018; Hamilton et al., 2018; Wang et al., 2018; Xiong et al., 2017; Welbl et al., 2018; Katsaklis et al., 2018), where either the model is provided with a knowledge graph to perform inference over or where the model must infer a knowledge graph from the text itself. However, unlike previous benchmarks in this domain—which are generally *transductive* and focus on leveraging and extracting knowledge graphs as a source of background knowledge about a fixed set of entities—CLUTRR requires *inductive logical reasoning*, where every example requires reasoning over a new set of previously unseen entities.

## 3 Benchmark Design

In order to design an NLU benchmark that explicitly tests inductive reasoning and systematic generalization, we build upon the classic ILP task of inferring family (i.e., kinship) relations (Hinton et al., 1986; Muggleton, 1991; Lavrac and Dzeroski, 1994; Kok and Domingos, 2007; Rocktäschel and Riedel, 2017). For example, given the facts that “Alice is Bob’s mother” and “Jim is Alice’s father”, one can infer with reasonable certainty that “Jim is Bob’s grandfather.” While this example may appear trivial, it is a challenging task to design models that can learn from data to *induce* the logical rules necessary to make such inferences, and it is even more challenging to design models that can systematically generalize by composing these induced rules.

Inspired by this classic task of logical induction and reasoning, the CLUTRR benchmark requires an NLU system to infer and reason about kinship

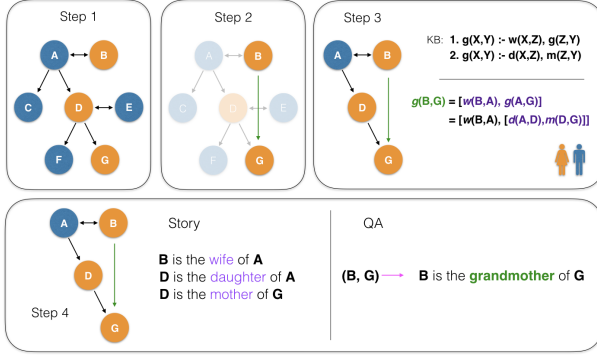


Figure 2: Data generation pipeline. Step 1: generate a kinship graph. Step 2: sample a target fact. Step 3: Use backward chaining to sample a set of facts. Step 4: Convert sampled facts to a natural language story.

relations by reading short stories. Requiring that the models learn directly from natural language makes this task much more challenging than the purely symbolic ILP setting. However, we leverage insights from traditional ILP to generate these stories in a semi-synthetic manner, providing precise control over the complexity of the reasoning required to solve the task.

In its entirety, the CLUTRR benchmark suite allows researchers to generate diverse semi-synthetic short stories to test different aspects of inductive reasoning capabilities. We publicly release the entire benchmark suite, including code to generate the semi-synthetic examples, the specific datasets that we introduce here, and the different baselines that we compare with.<sup>1</sup>

### 3.1 Overview of data generation process

The core idea behind the CLUTRR benchmark suite is the following: Given a natural language story describing a set of kinship relations, the goal is to infer the relationship between two entities, whose relationship is *not* explicitly stated in the story. To generate these stories, we first design a knowledge base (KB) with rules specifying how kinship relations resolve, and we use the following steps to create semi-synthetic stories based on this knowledge base:

**Step 1. Generate** a random kinship graph that satisfies the rules in our KB.

**Step 2. Sample a target fact** (i.e., relation) to predict from the kinship graph.

<sup>1</sup>Benchmark suite code can be obtained from <https://github.com/facebookresearch/clutrr>. Generated datasets are available to view in [this link](#).

**Step 3. Apply backward chaining** to sample a set of facts that can prove the target relation (and optionally sample a set of “distracting” or “irrelevant” noise facts).

**Step 4. Convert the sampled facts into a natural language story** through pre-specified text templates and crowd-sourced paraphrasing.

Figure 2 provides a high-level overview of this idea, and the following subsections describe the data generation process in detail, as well as the diagnostic flexibility afforded by CLUTRR.

### 3.2 Story generation

The short stories in CLUTRR are essentially narrativized renderings of a set of logical facts. In this section, we describe how we sample the logical facts that make up a story by generating random kinship graphs and using backward chaining to produce logical reasoning chains. The conversion from logical facts to natural language narratives is then described in Section 3.3.

**Terminology and background.** Following standard practice in formal semantics, we use the term *atom* to refer to a *predicate* symbol and a list of terms, such as  $[grandfatherOf, X, Y]$ , where the predicate `grandfatherOf` denotes the *relation* between the two *variables*,  $X$  and  $Y$ . We restrict the predicates to have an arity of 2, i.e., binary predicates. A logical *rule* in this setting is of the form  $\mathcal{H} \vdash \mathcal{B}$ , where  $\mathcal{B}$  is the *body* of the rule, i.e., a conjunction of two *atoms* ( $[\alpha_1, \alpha_2]$ ) and  $\mathcal{H}$  is the *head*, i.e., a single atom ( $\alpha$ ) that can be viewed as the goal or query. For instance, given a knowledge base (KB)  $R$  that contains the single rule  $[grandfatherOf, X, Y] \vdash [[fatherOf, X, Z], [fatherOf, Z, Y]]$ , the query  $[grandfatherOf, X, Y]$  evaluates to true if and only if the body  $\mathcal{B} = [[fatherOf, X, Z], [fatherOf, Z, Y]]$  is also true in a given world. A rule is called a *grounded* rule if all atoms in the rule are themselves *grounded*, i.e., all variables are replaced with *constants* or entities in a world. A *fact* is a grounded binary predicate. A *clause* is a conjunction of two or more atoms ( $\mathcal{C} = (\mathcal{H}_C \vdash \mathcal{B}_C = ([\alpha_1, \dots, \alpha_n]))$ ) which can be built using a set of rules.

In the context of our data generation process, we distinguish between the knowledge base,  $R$ , which contains a finite number of predicates and rules specifying how kinship relations in a family resolve, and a particular kinship graph  $G$ , which